TECHNICAL PROJECT · GROUP 2 · MSC CYBERSECURITY AND DATA SCIENCE, ESAIP ANGERS

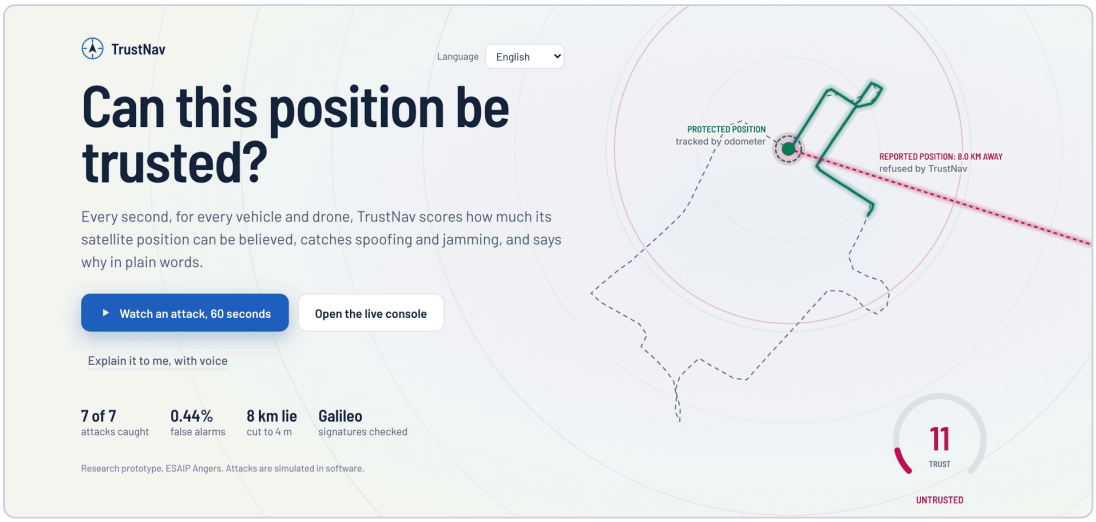
TrustNav: the demonstration in two minutes

The demonstration is online. This guide shows the four screens worth opening, and what each one proves.

> **trustnav.vercel.app** Nothing to install, no account. It works on a laptop and on a phone.

1 Open the link and click “Watch an attack, 60 seconds”

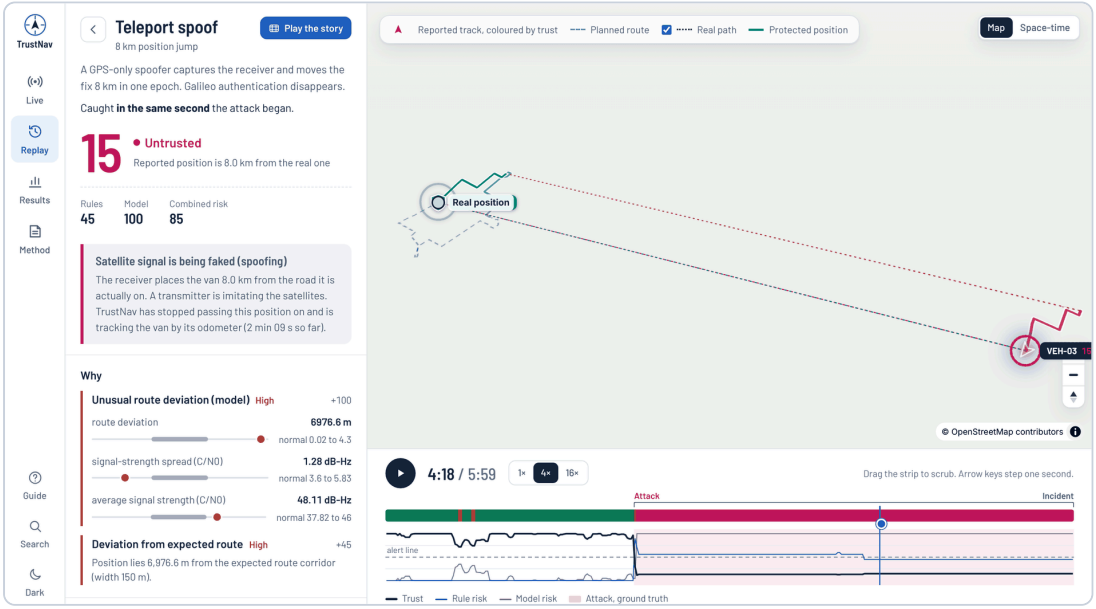
A van drives through Angers. A spoofer throws its reported position 8 km away. The trust score falls from the nineties to under 20, the false position is refused, and the van is tracked by its odometer. The story then plays by itself, with captions.



The opening screen during the attack: the reported position 8 km away is refused, the protected position follows the road.

2 Replay, Teleport spoof: what the operator reads

The score, the verdict in plain words, and the evidence: each unusual measure with its value against its normal range. On the map: the real path, the reported position 8 km away, and the dashed lie between them.



The replay at the attack. On the live site the streets of Angers appear under the tracks.

3 Results, Real attacks: beyond the simulator

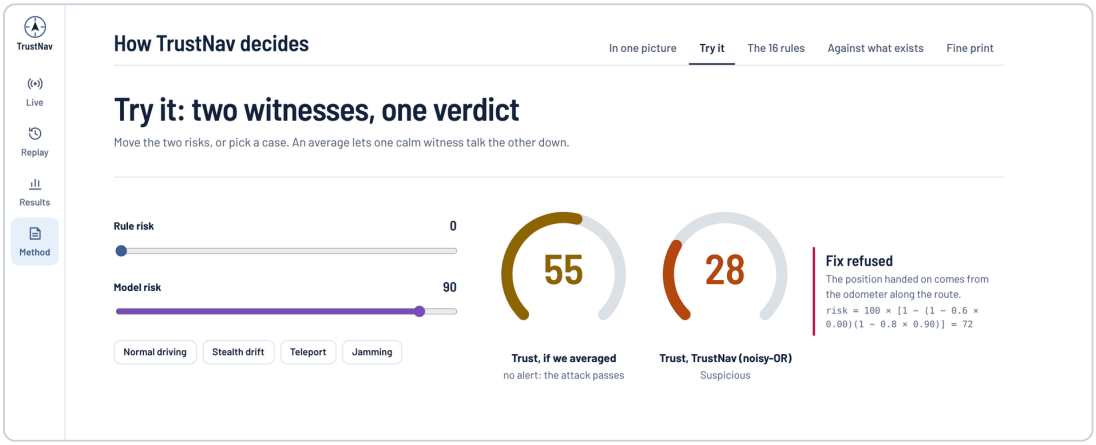
The detector replayed on public recordings of real attacks (Jammertest 2024, Norway). Our first version missed a phone captured by a real spoofer for 21 min 49 s. With a sixteenth rule, receiver time checked against the phone's own clock, the alert comes in the same second and 909 of 909 captured positions are refused.



What the phone reports ends 10.2 km off; what TrustNav hands on stays within 45 m.

4 Method, Try it: why two witnesses

Two sliders: the risk seen by the rules and the risk seen by the model. An average lets one calm witness talk the other down; the fusion we use does not. This is the design choice behind the score.



A model at 90 with silent rules: an average would let the attack pass, the noisy-OR refuses the fix.

Also worth a look

Results, Overview	99.3 % precision, 96.1 % recall and 0.44 % false alarms, on routes the model never saw in training
Results, Each attack	Ten recorded runs: when the alert came, what an ordinary screen would have shown, what TrustNav hands on
Method, The 16 rules	Every rule with its threshold and its weight
Source code	github.com/domail2karandeshmukh-cyber/trustnav: engine, cloud side, console, tests, paper and report

The paper (six pages, in English and in French), the report (44 pages), a six-page summary and the slides are sent with this guide. Our own attacks are simulated in software and the real ones are recordings made by others: nothing is ever transmitted.